

■IDENTIFICATION AND AUTHENTICATION POLICY

Organization: Blue Ridge Technology Solutions LLC

Control: 3.5.4 - Replay-Resistant Authentication

Version: 1.2 | Effective: 2026-01-15

Owner: Nathan Cruz, IT Manager | Approved: Victor Osei, Director of Operations

1. PURPOSE

This policy requires Blue Ridge Technology Solutions LLC to employ replay-resistant authentication mechanisms, per NIST SP 800-171 Rev. 2 control 3.5.4.

2. SCOPE

All authentication mechanisms used to access Blue Ridge Technology Solutions LLC CUI systems.

3. REQUIREMENTS

- a. Authentication protocols use challenge-response or cryptographic nonce mechanisms.
- b. Kerberos v5 is used for domain authentication (replay protection via timestamps and nonces).
- c. Azure AD authentication uses OAuth 2.0 tokens with expiration and nonce.
- d. NTLM v1 is disabled; NTLMv2 may be used only where Kerberos is not supported.
- e. Authentication tokens are short-lived and cannot be replayed after expiration.

4. RESPONSIBILITY

Nathan Cruz, IT Manager, configures authentication protocols with replay resistance.

5. ENFORCEMENT

Authentication protocols without replay protection are prohibited.

Signed: Victor Osei, Director of Operations | Blue Ridge Technology Solutions LLC | 2026-01-15